



Traffic Accident Risk Prediction Based on Multi-Granularity Spatio-Temporal Feature Decoupling and Recoupling Learning

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Abstract. Accurate prediction of traffic accident risks is crucial for enhancing urban traffic safety and the operational efficiency of road networks. However, traffic accident data typically suffers from extreme spatio-temporal sparsity including the zero-inflation phenomenon. Existing deep spatio-temporal networks are prone to feature entanglement and dependency degradation, which severely limit their predictive performance. To address these issues, this paper proposes a Multi-Granularity Spatio-Temporal Decoupled-Recoupled Network (MDR-Net) tailored for highly sparse data scenarios. Specifically, the model constructs a hierarchical spatio-temporal architecture that integrates both fine-grained and coarse-grained views. A cross-scale fusion module is introduced to facilitate feature interaction and complementarity across different granularities, while a weighted loss function is simultaneously incorporated to mitigate zero-inflation bias. Furthermore, MDR-Net employs a decoupled learning structure, utilizing independent spatial and temporal networks to extract corresponding features. Subsequently, a self-distillation-based recoupling mechanism is introduced to explicitly align feature distributions and reconstruct deep spatio-temporal dependencies. Comprehensive trials on two real-world datasets collected from New York and Chicago reveal that MDR-Net achieves substantial improvements over current leading baseline methods.

Keywords: Traffic accident risk prediction · Spatiotemporal data · Zero inflation · Multigranularity learning · Disentangled representation learning

1 Introduction

Urban traffic safety is central to smart city governance, with accidents causing millions of casualties and a severe socioeconomic burden annually [1]. The widespread adoption of IoT and mobile computing has generated massive spatiotemporal data, providing crucial support for precise risk prediction and public safety. Traffic accident risk prediction inherently involves spatiotemporal sequence forecasting. Early methods progressed from cascaded networks to joint spatiotemporal architectures, but tightly coupled models (e.g., Hetero-ConvLSTM [2]) suffered from feature entanglement due

to redundant noise, while recent decoupled paradigms (e.g., STDIL [3]) risk degrading coupled dependencies under extreme data sparsity, hindering the reconstruction of long-range correlations. Although weighted losses [4] offer partial relief, they operate mainly at the optimization or output level without enhancing sparse-signal representation at the feature level. Multi-granularity feature construction (e.g., CA-STNet [5]) mitigates sparsity but often relies on simple concatenation or isolated processing, lacking deep cross-scale interactions and struggling to capture consistency between local details and global semantics. To handle data sparsity and complex spatiotemporal correlations, MDR-Net uses multi-granularity feature decoupling and recoupling. Its main contributions are: (1) a hierarchical multi-granularity framework with cross-scale fusion and weighted loss to reduce zero inflation; (2) a decoupled spatial-temporal architecture with self-distillation recoupling to model deep dependencies; (3) extensive experiments on New York and Chicago datasets demonstrating significant gains over baselines, with ablation validating all modules.

2 Related Work

2.1 Statistical and Machine Learning Methods for Traffic Risk Prediction

Statistical models such as ARIMA are robust in short-term prediction [6]. Ihueze et al. [7] extended ARIMA to ARIMAX by incorporating human, vehicle, and environmental factors to forecast regional accident frequencies. For real-time risk prediction, Hossain et al. [8] first applied Bayesian networks; Sun et al. [9] later introduced a dynamic Bayesian architecture with parameter learning to better handle complex traffic flows. In post-accident analysis, Hamad et al. [10] employed Random Forests to estimate accident durations and reduce long-span prediction errors, and Tirtha et al. [11] developed a Copula-based Logit joint model to examine how multi-source external variables jointly influence accident type and duration.

2.2 Deep Learning-Based Traffic Risk Prediction

Early deep learning methods used convolutional networks or RNNs for single-dimensional feature extraction; hybrid models like ConvLSTM and TWCCnet [12] later captured spatio-temporal dependencies jointly but assumed regular Euclidean spaces, limiting adaptability to non-Euclidean road topologies. Graph Convolutional Networks were therefore introduced: DSTGCN [13] built static graphs from physical road networks, yet these static structures fail to represent dynamic traffic states. To overcome this, GSNet [4] jointly modeled geographical proximity and semantic similarity (e.g., POIs) for dynamic multi-view fusion. As prediction granularity became finer, extreme data sparsity (zero-inflation) emerged as a key bottleneck. RiskOracle [14] introduced differential time-varying graphs and multi-scale risk co-learning, while MGHSTN [15] employed multi-granularity hierarchical networks and temporal attention for cross-scale knowledge transfer. Nevertheless, deep cross-scale interaction of multi-granularity features and the avoidance of dependency degradation after spatio-temporal decoupling under sparse data remain underexplored. To this end, this paper proposes a network

that integrates multi-granularity feature learning with a decoupling-recoupling mechanism, aiming to enhance sparse data representation at the feature level and explicitly reconstruct deep spatio-temporal dependencies.

3 Methodology

3.1 Overall Architecture

As illustrated in Fig. 1, the proposed MDR-Net primarily consists of five components. These include a spatial network, a temporal network, a spatio-temporal recoupling mechanism, a cross-scale interaction and fusion module, and a prediction module. Specifically, the spatial network is composed of a mapping layer and multiple spatial blocks, where each spatial block contains a multi-scale spatial convolution module and a dual attention module. The temporal network is formed by stacking multiple temporal blocks, with each temporal block including a multi-scale temporal convolution and an attention-based Transformer. Through the synergy of these components, the model achieves the decoupling and recoupling of spatio-temporal features, ultimately outputting the predicted traffic accident risk values for each region.

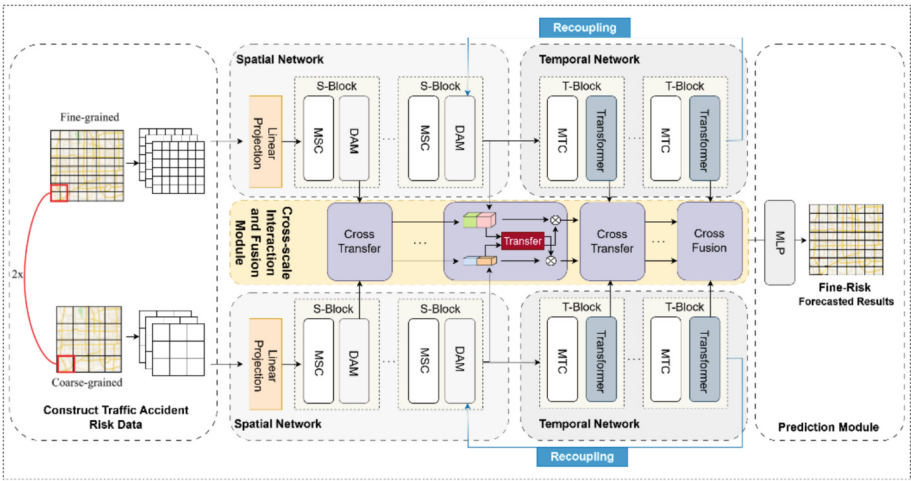


Fig. 1. Overall architecture of the proposed MDR-Net

3.2 Spatial Network

The spatial network consists of a linear projection and multiple spatial blocks, as illustrated in Fig. 1. Specifically, it comprises (1) a Linear Projection module and (2) spatial blocks, with each consisting of a Multi-Scale Spatial Convolution (MSC) module and a Dual Attention Module (DAM).

Linear Projection. In this paper, a linear projection transformation is applied to the input features to map them into a feature representation that is easier for the model to comprehend. The transformation process is formulated as Eq. (1).

$$H_S^l = W_l H_S^{in} \quad (1)$$

where $H_S^{in} \in \mathbb{R}^{C \times T \times N}$ is the input feature matrix, and W_l denotes a learnable parameter matrix.

Multi-Scale Spatial Convolution (MSC). This paper utilizes 3D convolutional kernels ($1 \times 3 \times 3$, $1 \times 5 \times 5$, and $1 \times 7 \times 7$) with the temporal dimension fixed to 1 to extract spatial features across different scales. Specifically, local spatial features are extracted through smaller convolutional kernels, while broader contextual information is captured by larger kernels. Combined with average pooling, this enables the model to perform multi-scale spatial feature extraction from the input data. Feature maps produced by convolutions at different scales are first concatenated along the channel axis, then passed through a $1 \times 1 \times 1$ convolution to form the final multi-scale representation. This design effectively aggregates information across scales, strengthening the model's ability to capture complex spatial correlations. The computation is formulated as follows.

$$H_S^k = \text{MaxPool}\left(\text{CNN}_{1 \times k \times k}\left(H_S^{in}, W_S^k\right)\right), \quad k = 3, 5, 7 \quad (2)$$

$$H_S^{\text{MSC}} = \sigma\left(W * \text{Cat}\left(H_S^3, H_S^5, H_S^7\right) + b\right) \quad (3)$$

where $\text{CNN}_{1 \times k \times k}(\cdot, W_S^k)$ represents the multi-scale spatial convolution with a kernel size of $1 \times k \times k$, MaxPool is the max pooling layer, Cat denotes the concatenation operation, $*$ represents the convolution operation, σ is the GELU activation function, and W and b are learnable parameters.

Dual Attention Module (DAM). A dual attention mechanism is designed to perform feature recalibration across the channel and spatial dimensions. This directs the model's emphasis toward high-risk locations and informative channels, thereby strengthening its spatial feature representations.

Channel Attention. To strengthen the traffic accident risk features along the channel direction, this paper directly applies matrix multiplication to the original feature $H \in \mathbb{R}^{C \times H \times W}$ to compute the attention matrix $A \in \mathbb{R}^{C \times C}$.

First, the input feature $H \in \mathbb{R}^{C \times H \times W}$ is reshaped to obtain the feature in $\mathbb{R}^{C \times N}$, where $N = H \times W$. Then, matrix multiplication is performed between this feature and its transpose, followed by a softmax function to obtain the channel attention matrix A_C .

$$A_C = \text{softmax}\left(H \cdot H^T\right) \quad (4)$$

Next, matrix multiplication is performed between the channel attention matrix A_C and the reshaped feature H , and a residual connection is introduced to obtain the final output feature enhanced by the channel. This process is formulated as follows.

$$O_C = H * A_C + H \quad (5)$$

where H is the input feature, H^T is the transpose of the input feature, $*$ denotes element-wise multiplication, A_C is the channel attention matrix, and O_C is the channel attention feature output enhanced by the channel attention.

Spatial Attention. This module enhances spatial features in a symmetric manner to channel attention. Specifically, the input feature $H \in \mathbb{R}^{C \times H \times W}$ is first reshaped to obtain the feature $H \in \mathbb{R}^{N \times C}$, where $N = H \times W$. The spatial attention matrix $A_S \in \mathbb{R}^{N \times N}$ and the enhanced feature O_S are then obtained analogously via:

$$A_S = \text{softmax}(H \cdot H^T) \quad (6)$$

$$O_S = H * A_S + H \quad (7)$$

To better fuse O_C and O_S , this paper utilizes an adaptive fusion matrix to guide their fusion. This process is described as follows.

$$O_{\text{spatial}} = W_R \odot (O_C + O_S) \quad (8)$$

where $W_R \in \mathbb{R}^{C \times T \times N}$ is the spatio-temporal recoupling matrix, and $\odot$ represents element-wise multiplication, which is the key to achieving the subsequent spatio-temporal recoupling.

3.3 Temporal Network

The temporal network stacks multiple temporal blocks, each comprising a multi-scale temporal convolution (MTC) module and a Transformer. To reduce the high cost of standard global self-attention, we adopt an efficient k-NN Attention variant [16]. The computational details of both mechanisms are given below.

Multi-Scale Temporal Convolution (MTC). We employ 3D convolutional kernels of different sizes ($3 \times 1 \times 1$, $5 \times 1 \times 1$, $7 \times 1 \times 1$) to capture locally refined temporal features, enabling the model to focus on short-term temporal correlations. Assuming the input temporal feature is $H_T^{\text{in}} \in \mathbb{R}^{C \times T \times N}$, the process is formulated as Eqs. (9) and (10).

$$H_T^k = \text{MaxPool}\left(\text{CNN}_{k \times 1 \times 1}\left(H_T^{\text{in}}, W_T^k\right)\right), \quad k = 3, 5, 7 \quad (9)$$

$$H_T^{\text{MTC}} = \sigma\left(W * \text{Cat}\left(H_T^3, H_T^5\right) + b\right) \quad (10)$$

where $\text{CNN}_{k \times 1 \times 1}(\cdot, W_T^k)$ represents the multi-scale temporal convolution, W_T^k is a learnable parameter, Cat represents the concatenation operation, σ is the GELU activation function, and W and b are learnable parameters.

k-NN Based Transformer. To accelerate training, this paper adopts k-NN Attention to replace the global attention, computing the attention matrix by selecting the top- k most similar key-value pairs. Specifically, this process can be described as Eq. (11).

$$O_{\text{Temporal}} = \text{FFN}\left(\text{Att}_{\text{kNN}}\left(H_T^{\text{MTC}} + W^P\right)\right) + H_T^{\text{MTC}} \quad (11)$$

where $\text{Att}_{\text{kNN}}(\cdot)$ represents the multi-head k-NN self-attention (in this paper, we select the top 70% of keys and values to compute the attention matrix), FFN is the feed-forward neural network, and W^P is the position embedding, which is a trainable parameter. Simultaneously, residual connections are employed to ensure training stability and accelerate the training of the model.

3.4 Spatio-Temporal Recoupling Mechanism

We propose a spatio-temporal recoupling mechanism that uses a self-distillation loss $\text{Loss}_{\text{distill}}$ to transfer correlation knowledge from the temporal network to the spatial network, thereby reconstructing deep dependencies. The computation is given in Eq. (12).

$$\text{Loss}_{\text{distill}} = L_{\text{distill}}(W_T, W_R) \quad \text{s.t. } L_{\text{distill}}(x, y) = \frac{1}{N} \sum_{i=1}^N KL(x_i/\tau - y_i/\tau) \quad (12)$$

where N is the batch size, KL represents the KL divergence, τ is the distillation temperature coefficient, W_R is the spatio-temporal recoupling matrix, and $W_T = \text{Softmax}(O_{\text{Temporal}})$.

3.5 Cross-Scale Interaction and Fusion Module

We design a cross-scale interaction and fusion mechanism that creates a dedicated interactive feature flow between the coarse-grained grids and fine-grained grids, achieving deep filtering and complementary fusion of multi-scale features.

This paper defines the coarse-fine granularity feature transformation matrix $R_{fc} \in \mathbb{R}^{n \times N}$, where $n = N/k^2$ and $k = 2$. This matrix describes the two types of grids as shown in Eq. (13).

$$R_{fc}(i, j) = \begin{cases} 1, & \text{if } r_i^F \in r_j^C \\ 0, & \text{otherwise} \end{cases} \quad (13)$$

where r_i^F represents the i -th grid in the coarse-grained grid, and r_j^C represents the j -th grid in the fine-grained grid. To learn the complex correlations between regions, this paper proposes a cross-scale interaction and fusion method, which conducts cross-scale information interaction within two different scale spaces (coarse and fine granularities) to achieve feature filtering and fusion. The specific details are as follows. We assume the coarse-grained input feature is $H_c \in \mathbb{R}^{C \times T \times n}$ and the fine-grained input feature is $H_f \in \mathbb{R}^{C \times T \times N}$.

$$H_f^{\text{out}} = W_1 * \text{Concat}(H_f, (H_c R_{fc} W_c)) + b_1 \quad (13)$$

$$H_c^{\text{out}} = W_2 * \text{Concat}(H_c, (H_f R_{fc}^T W_f)) + b_2 \quad (13)$$

where $*$ represents the convolution operation, R_{fc} is the transformation matrix, and R_{fc}^T is its transpose. The variables W_c and W_f are trainable parameter matrices, and W_1 , W_2 , b_1 , and b_2 are trainable parameters.

3.6 Risk Prediction Module

Assuming the spatio-temporal feature after cross-scale fusion is denoted as H_{ST}^{fusion} , the traffic accident risk prediction result $\tilde{\mathbb{Y}}$ can be expressed as Eq. (16).

$$\tilde{\mathbb{Y}} = W_2 * \sigma(W_1 * H_{ST}^{\text{fusion}} + b_1) + b_2 \quad (16)$$

where W_1 , W_2 , b_1 , and b_2 are learnable parameters, and $\sigma(\cdot)$ represents the ReLU activation function.

To mitigate the zero-inflation issue in traffic accident data, a weighted RMSE loss function is adopted. Samples are grouped into four risk tiers according to their risk values, taken from the set $\{0, 1, 2, 3\}$ and denoted by θ . During loss computation, distinct weights are assigned to samples $\mathbb{Y}(i)$ belonging to each tier. The loss is formulated as follows.

$$\text{Loss}(\mathbb{Y}, \tilde{\mathbb{Y}}) = \frac{1}{2} \sum_{i \in \theta} \omega_i \left(\mathbb{Y}(i) - \tilde{\mathbb{Y}}(i) \right)^2 \quad (17)$$

where $\mathbb{Y}$ represents the ground-truth risk value, $\tilde{\mathbb{Y}}$ represents the predicted risk value, and ω_i is the weight corresponding to the traffic accident risk value of different levels. The final optimization objective of the model is the sum of the prediction loss and the distillation loss.

$$\text{Loss} = \text{Loss}(\mathbb{Y}, \tilde{\mathbb{Y}}) + \text{Loss}_{\text{distill}} \quad (18)$$

4 Experiments

4.1 Datasets

In this paper, we conduct experiments using two real-world traffic datasets. These are the New York City (NYC) dataset and the Chicago dataset. The NYC dataset contains records of traffic accidents, taxi trips, POI (Point of Interest), and weather information for the entire year of 2013. The Chicago dataset covers traffic accidents, taxi trips, and weather data from February to September 2016.

4.2 Experimental Settings

We implement MDR-Net in PyTorch and train it on an NVIDIA RTX 4060 (8GB) GPU. The city area is partitioned into 400 fine-grained grids, and all inputs are min-max normalized to $[0, 1]$. We chronologically split the data into training, validation, and test sets (6:2:2). The initial learning rate is 1×10^{-4} . For the weighted RMSE, risk levels receive weights of $\{0.05, 0.2, 0.25, 0.5\}$. All reported results are averaged over five independent runs.

4.3 Evaluation Metrics

As shown in Eqs. (19)–(21), this paper employs RMSE to measure the global regression error and utilizes Recall and MAP to evaluate the ranking accuracy of high-risk areas.

$$\text{RMSE} = \sqrt{\frac{1}{T} \sum_{t=1}^T \|X_t - \hat{X}_t\|^2} \quad (19)$$

$$\text{Recall} = \frac{1}{T} \sum_{t=1}^T \frac{|G_t \cap R_t|}{|R_t|} \quad (20)$$

$$\text{MAP} = \frac{1}{T} \sum_{t=1}^T \frac{\sum_{j=1}^{|R_t|} \text{pre}(j) \times \text{recall}(j)}{|R_t|} \quad (21)$$

where X_t represents the ground-truth traffic accident risk of all regions at time step t , while $\hat{X}_t$ denotes the corresponding predicted values. R_t is the set of grid regions where traffic accidents actually occurred during time slot t , and G_t is the set of predicted high-risk accident regions at time slot t . $\text{pre}(j)$ represents the precision at rank j for the top- j high-risk regions, and $\text{recall}(j)$ denotes the recall rate for region j .

4.4 Baseline Models

We compare MDR-Net with eight well-acknowledged baselines, including (1) classical statistical and machine learning methods: HA, MLP, and GRU [17]; (2) convolutional and recurrent hybrid architectures: SDCAE [18], ConvLSTM [19], and 3D-CANet [20]; and (3) graph-enhanced spatiotemporal methods: GSNet and TWCCNet [21].

4.5 Performance Experiments

As shown in Table 1, MDR-Net achieves superior performance on both datasets. Single-dimension methods (HA, MLP, GRU, SDCAE) fail to capture complex spatiotemporal coupling; ConvLSTM is limited by local receptive fields and struggles with global correlations; while GSNet and TWCCNet, despite modeling geographical and semantic dependencies, lose deep spatiotemporal correlations due to their tightly entangled architectures. In contrast, MDR-Net alleviates sparsity via multi-granularity joint modeling, explicitly reconstructs deep spatiotemporal dependencies through its decoupling–recoupling mechanism with self-distillation, and further suppresses zero-inflation bias using a weighted loss, leading to substantially improved overall risk prediction.

Table 1. Performance comparison of different methods. Best results are in bold and second-best are underlined.

Models	Chicago		NYC			
	RMSE	Recall	MAP	RMSE	Recall	MAP
HA	14.9582	13.81%	0.0573	10.3244	24.43%	0.1048
MLP	12.5116	17.52%	0.0632	8.4288	27.28%	0.1196
GRU	12.6482	17.84%	0.0664	8.3376	28.09%	0.1228
ConvLSTM	11.1308	18.85%	0.0789	7.9506	30.99%	0.1526
SDCAE	11.3383	18.79%	0.0753	7.9775	30.81%	0.1594
3D-CANet	10.7654	20.32%	0.0854	7.6051	32.85%	0.1684
GSNet	<u>10.5965</u>	19.92%	0.0868	7.5158	33.41%	0.1861
TWCCnet	10.734	<u>20.75%</u>	<u>0.0945</u>	<u>7.2758</u>	<u>33.89%</u>	<u>0.1893</u>
MDR-Net	10.4182	21.09%	0.1008	7.1364	34.27%	0.1907

4.6 Ablation Study

To further demonstrate the effectiveness of different components, we designed five variants on the NYC dataset: (i) -MSC (removing Multi-Scale Spatial Convolution), (ii) -MTC (removing Multi-Scale Temporal Convolution), (iii) -Transformer (removing the k-NN Transformer), (iv) -DAMR (removing Dual Attention and recoupling mechanism), and (v) -CSIF (removing Cross-Scale Interaction and Fusion).

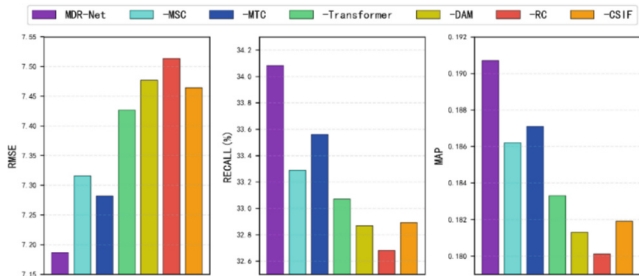


Fig. 2. Ablation study on the NYC dataset

Figure 2 shows that removing MSC or MTC, the Transformer, DAMR, or CSIF all caused clear performance degradation. The drops confirm the importance of spatial and short-term temporal correlations, the indispensability of deep temporal mining for the recoupling mechanism, the Dual Attention Module’s enhancement of channel-spatial features, and the need for cross-scale interaction and filtering to avoid collapsing into two isolated feature flows with only a simple final fusion.

4.7 Hyperparameter Experiments

We determine the optimal hyperparameters via grid search, selecting values that minimize RMSE on the validation set under a strict control-variable protocol.

Table 2. Kernel selection experiment for multi-scale temporal convolution.

Kernel Size (MTC)	NYC		
	RMSE	Recall	MAP
$3 \times 1 \times 1$	7.4472	33.05%	0.1812
$3 \times 1 \times 1 + 5 \times 1 \times 1$	7.3816	33.53%	0.1863
$5 \times 1 \times 1 + 7 \times 1 \times 1$	7.5309	32.69%	0.1798
$3 \times 1 \times 1 + 5 \times 1 \times 1 + 7 \times 1 \times 1$ (MTC)	7.1864	34.27%	0.1907

Table 2 reports the results of different temporal convolutional kernel scale configurations. The experiments show that any reduced single-scale or dual-scale combination leads to higher RMSE and lower Recall and MAP, confirming that a complete multi-scale setup is essential for effective initial mining of complex spatio-temporal correlations.

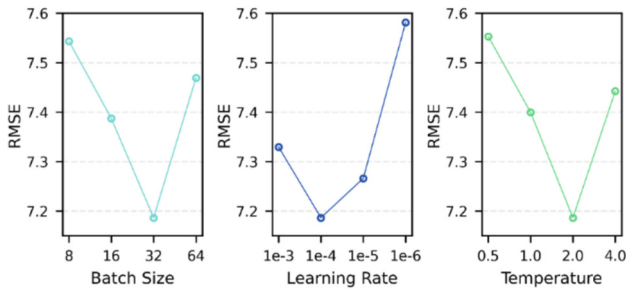


Fig. 3. RMSE comparison for hyperparameter selection

Figure 3 shows the sensitivity analysis of batch size, initial learning rate, and temperature τ on the NYC dataset. The lowest RMSE occurs at batch size 32 and learning rate 1×10^{-4} ; extreme values instead cause gradient oscillations, local optima, and weakened generalization. τ , which governs knowledge transfer intensity in the recoupling mechanism, critically influences spatio-temporal dependency reconstruction, with 2.0 being optimal.

5 Conclusion

This paper proposes a novel Spatio-Temporal Recoupling Network (MDR-Net) for traffic accident risk prediction, capable of more accurately forecasting fine-grained, area-specific traffic accident risks across an entire urban region. First, a two-layer spatio-temporal learning network is designed to capture the spatio-temporal correlations inherent in traffic accident risks. Second, a spatio-temporal recoupling mechanism is introduced to reconstruct spatio-temporal dependencies and optimize the model architecture through a self-distillation process. Subsequently, a cross-scale fusion module is designed to serve as a critical feature stream for filtering and fusing spatio-temporal features derived from both coarse-grained and fine-grained scales. Finally, rigorous experiments on two real-world traffic datasets from New York and Chicago confirm that MDR-Net delivers clear advantages in both predictive accuracy and computational efficiency. Future work will explore dynamic graph structure learning and adaptive hyperedge generation to capture time-varying, higher-order correlations in traffic risks.

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